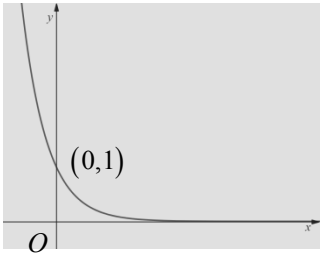
1 (a)	$y = 2x^2(x-5) = 2x^3 - 10x^2$	B1
	$\frac{dy}{dx} = 6x^2 - 20x$	M1
	Sets $\frac{dy}{dx} = 0 \Rightarrow 6x^2 - 20x = 0 \Rightarrow x = 0, \frac{10}{3}$ oe	dM1 A1 **(4)**
(b)	$x \leqslant 0, \quad x \geqslant \frac{10}{3}$	M1 A1 **(2)** **(6 marks)**
Alt (a)	$\left(\frac{d(uv)}{dx} = uv' + vu'\right) \qquad u = 2x^2, u' = 4x, v = x-5, v' = 1$	B1
	$\left(\frac{dy}{dx} =\right) Ax(x-5) + Bx^2$	M1
	Sets $\frac{dy}{dx} = 0 \Rightarrow 4x(x-5) + 2x^2 = 0 \Rightarrow x = 0, \frac{10}{3}$	dM1 A1

2 (a)	States or uses $r = 1.02$ Attempts $25000 \times 1.02^{13} = 32340$ or 32341 or 32300	B1 M1 A1 **(3)**
(b)	Attempts $\frac{a\left(r^n - 1\right)}{\left(r - 1\right)} = \frac{25000\left(1.02^{14} - 1\right)}{1.02 - 1}$ or $\frac{125000\left(1.02^{14} - 1\right)}{1.02 - 1}$ £1 997 000	M1 A1 A1 **(3)** **(6 marks)**

3. (a)	$\left(1+\frac{x}{4}\right)^{12} = 1+12\left(\frac{x}{4}\right)^{1}+\frac{12\times11}{2}\left(\frac{x}{4}\right)^{2}+\frac{12\times11\times10}{3!}\left(\frac{x}{4}\right)^{3}+\ldots$	M1
	$= 1+3x,+\frac{33}{8}x^{2}+\frac{55}{16}x^{3}$	B1, A1
		(3)
(b)	$\left(\frac{x^2+8}{x^5}\right)\left(1+\frac{x}{4}\right)^{12} =$	
	Sight of a term independent of x $= "\frac{55}{16}"$ or $8\times {}^{12}C_5\left(\frac{1}{4}\right)^5 \left(=\frac{99}{16}\right)$	B1ft,
	$"\frac{55}{16}" +8\times {}^{12}C_5\left(\frac{1}{4}\right)^5 = \frac{55}{16}+\frac{99}{16}=\frac{77}{8}$	M1 A1
		(3)
		(6 marks)

4 (a)	-35	B1 **(1)**
(b)	Attempts $f\left(\pm\frac{2}{3}\right)=0 \rightarrow \left(\frac{2}{3}-3\right)\left(3\left(\frac{2}{3}\right)^2+\left(\frac{2}{3}\right)+a\right)-35=0$	M1
	$-\frac{7}{3}(2+a)=35 \Rightarrow a=-17$ *	A1* **(2)**
(c)	$f(x)=(x-3)(3x^2+x-17)-35=3x^3-8x^2-20x+16$	B1
	$=(3x-2)(x^2-2x-8)$	M1 A1
	$=(3x-2)(x+2)(x-4)$	M1 A1 **(5)** **(8 marks)**

5.(a)



Shape or intercept at 1 — M1

Fully correct — A1

(2)

$h = 0.5$ — B1

(b) $\text{Area} \approx \frac{0.5}{2}\left\{4.25 + 16.06 + 2\times\left(6.427 + 9.125 + 12.34\right)\right\}$ — M1

$= \text{awrt } 19.0$ — A1

(3)

(c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) \mathrm{d}x = \int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x - 3x \right) \mathrm{d}x = (b) - \left[\frac{3}{2}x^2\right]_2^4$ — M1

$= \text{awrt } 1.0$ — A1ft

(2)

(7 marks)

6(a)	Attempts line with gradient -2 and point $(4,-1)$ $y+1=-2(x-4)$ $y=-2x+7$	M1 A1 **(2)**
(b)	$y=\frac{1}{2}x$ meets $y=-2x+7$ when $\frac{1}{2}x=-2x+7\Rightarrow x=\frac{14}{5}, y=\frac{7}{5}$ oe	M1 A1
	Attempts $r^2=\left(4-"\frac{14}{5}"\right)^2+\left(-1-"\frac{7}{5}"\right)^2=\frac{36}{5}$ oe	dM1 A1
	$(x-4)^2+(y+1)^2=\frac{36}{5}$ oe	A1 **(5)** **(7 marks)**

7. (i)	$\log_a\left(\frac{\sqrt{a}}{b}\right)=\frac{1}{2}\log_a a-\log_a b=\frac{1}{2}-k$	M1 A1 **(2)**
(ii)	$\frac{\log_a a^2 b}{\log_a b^3}=\frac{2\log_a a+\log_a b}{3\log_a b}=\frac{2+k}{3k}$	M1 A1 **(2)**
(iii)	$\sum_{n=1}^{50}\left(k+\log_a b^n\right)=50k+\left(1k+2k+3k+...+50k\right)$ or $\left(2k+3k+4k+...+51k\right)$	
	Uses the sum formula an AP with $n=50,\ d=k$	M1
	$S=50k+\frac{50}{2}\left(2k+49k\right)$ $\qquad$ $S=\frac{50}{2}\left(2k+51k\right)$	A1
	$=1325k$	A1 **(3)** **(7 marks)**

8 (i)	$\int \frac{8\sqrt{x}-5}{2x^2}\,\mathrm{d}x = \int 4x^{-\frac{3}{2}} - \frac{5}{2}x^{-2}\,\mathrm{d}x$	B1
	$= -8x^{-\frac{1}{2}} + \frac{5}{2}x^{-1} \quad (+C)$	M1 A1
	$\int_2^4 \left(4x^{-\frac{3}{2}} - \frac{5}{2}x^{-2}\right)\mathrm{d}x = \left(-4+\frac{5}{8}\right) - \left(-4\sqrt{2}+\frac{5}{4}\right) = 4\sqrt{2} - \frac{37}{8}$	dM1 A1
		(5)
(ii)	$\int \left(\frac{1}{2}x^2 + k\right)\mathrm{d}x = \left[\frac{1}{6}x^3 + kx\right]$	M1 A1
	$\int_{-3}^6 \left(\frac{1}{2}x^2 + k\right)\mathrm{d}x = 55 \Rightarrow \left[\frac{1}{6}x^3 + kx\right]_{-3}^6 = 55 \Rightarrow (36+6k) - \left(-\frac{9}{2}-3k\right) = 55 \Rightarrow k = \ldots$	dM1
	$k = \frac{29}{18}$	A1
		(4)
		(9 marks)

9.(i)	$3\sin(2\theta-10°)=1\Rightarrow(2\theta-10°)=\arcsin\left(\frac{1}{3}\right)$	M1
	$\theta=\frac{19.47+10}{2},\frac{160.53+10}{2}$	dM1
	$\theta=$ awrt $14.7°, 85.3°$	A1, A1
		(4)
(ii) (a)	Writes $\frac{1}{\tan\alpha}-\sin\alpha=2\sin\alpha-\frac{1}{\tan\alpha}$ oe	M1
	$\frac{2}{\tan\alpha}=3\sin\alpha\Rightarrow\frac{2\cos\alpha}{\sin\alpha}=3\sin\alpha\Rightarrow2\cos\alpha=3\sin^2\alpha$ *	dM1 A1*
		(3)
(b)	$2\cos\alpha=3\sin^2\alpha\Rightarrow2\cos\alpha=3(1-\cos^2\alpha)$	M1
	$3\cos^2\alpha+2\cos\alpha-3=0$	A1
	Attempts to solve $3\cos^2\alpha+2\cos\alpha-3=0\Rightarrow\cos\alpha=\frac{-2\pm\sqrt{40}}{6}$ oe	dM1 A1
	$\alpha=5.517$ radians	A1
		(5)
		(12 marks)

10.(a)	$x=2, y=5 \Rightarrow 5=8a-12+6+b$	M1
	$\frac{dy}{dx}=3ax^2-6x+3$ **AND** $x=2, \frac{dy}{dx}=7 \Rightarrow 7=12a-12+3$	M1
	Solves $11=8a+b$ and $7=12a-9 \Rightarrow a=\frac{4}{3}, b=\frac{1}{3}$	A1 A1
		(4)
(b)	Sets $\frac{dy}{dx}=3ax^2-6x+3=0$ with their value of a and b	M1
	$4x^2-6x+3=0$ and attempts "b^2-4ac"	dM1
	$b^2-4ac=-12<0$ hence there are no turning points oe	A1*
		(3)
		(7 marks)